

SEC P vs NP log 2026-05-17

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-17 03:32 UTC

SEC P vs NP — daily log 2026-05-17

9 cycles recorded

GCT Lie-Stabilizer Codimension Linearly Bounds ACC^0 Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Geometric Complexity Theory: Lie-algebra stabilizers of multilinear polynomial extensions in the Mulmuley-Sohoni-Bürgisser-Ikenmeyer framework $\times \text{ACC}^0[m]$ circuit size for Boolean functions (Sipser-like targets)
- **Recorded:** 2026-05-17 00:01 UTC
- **Entry ID:** 83d26c2ef069

Statement

For a Boolean function $f: \{-1, +1\}^n \rightarrow \{-1, +1\}$, let $F_f(x) = \sum_{S \subseteq [n]} \hat{f}(S) \cdot \prod_{i \in S} x_i$ be its multilinear Walsh extension over $\mathbb{Q}$, let $L(F_f) := \{A \in \mathfrak{gl}_n(\mathbb{Q}) : \sum_{i,j} A_{ij} x_j \partial_i F_f \equiv 0 \text{ in } \mathbb{Q}[x_1, \dots, x_n]\}$ be its GCT Lie stabilizer, and set $\gamma(f) := n^2 - \dim_{\mathbb{Q}} L(F_f)$. Conjecture: there is an absolute constant $C \leq 10$ such that every $\text{ACC}^0[m]$ circuit of size s computing f satisfies $\gamma(f) \geq n^2 - C \cdot s$. A single $\text{ACC}^0[m]$ circuit of size s computing some f with $\gamma(f) < n^2 - 10 \cdot s$ falsifies the conjecture.

Rationale

GCT's symmetry-characterization principle says polynomials whose stabilizer Lie algebras exceed generic dimension are atypical; we invert this to claim small ACC^0 circuits cannot accidentally synthesize multilinear extensions of anomalously high symmetry. The Lie stabilizer is GL_n -equivariant rather than truth-table-dense, so it dodges Razborov-Rudich's 'large + constructive' clause, and it is computed from the GL_n action on polynomials, not from black-box algebraic extensions — placing the invariant outside the algebrization and relativization regimes that GCT was designed to escape.

Novelty

- Judge: NOVEL over 11 arXiv hits

Top hits consulted: - [1212.5380v2] On properties of principal elements of Frobenius Lie algebras - [1606.05050v1] Proof Complexity Lower Bounds from Algebraic Circuit Complexity - [2411.11095v3] Invariant theory and coefficient algebras of Lie algebras - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [1504.04131v2] Formulas for the Walsh coefficients of smooth functions and their application to bounds on the Walsh coefficients

Empirical Test

- exit code: 1, elapsed: 0.11s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_698adb63.py", line 102, in <module>
    result = run_trial(seed)
    ~~~~~^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_698adb63.py", line 86, in run_trial
    gamma = gaussian_elimination(system)
    ~~~~~^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_698adb63.py", line 65, in gaussian_elim
    if matrix[i][i] == 0:
    ~~~~~^
```

IndexError: list index out of range

Judge reasoning

The test crashed with an `IndexError` in `gaussian_elimination` before any of the 480 trials or reference-function checks produced slack data, so the pre-registered support condition cannot be evaluated. Per Rule 2, a crash yields **INCONCLUSIVE**. | next: Fix the Gaussian elimination routine (guard against out-of-range pivot indexing and handle rectangular/short matrices) and rerun the full (n,s)-grid plus reference-function suite to obtain actual slack statistics.

2-adic Newton Slopes of DISJ Sub-blocks Scale as $\sqrt{n}$

- **Verdict:** INCONCLUSIVE
- **Bridge:** p-adic analysis (Newton polygons over $\mathbb{Q}_2$ of integer characteristic polynomials) $\times$ Randomized communication complexity of DISJOINTNESS via average-case tensor invariants on sub-blocks
- **Recorded:** 2026-05-17 00:26 UTC
- **Entry ID:** efd166108501

Statement

Let $M_n \in \{-1, +1\}^{\{2^n \times 2^n\}}$ be the signed disjointness matrix $M_n[x, y] = (-1)^{|x \cap y|}$ where x, y are interpreted as subsets of $[n]$. For each n , draw uniformly random row set R and column set C with $|R|=|C|=k=\min(2n, 40)$, form $A := M_n[R, C]$, compute its exact integer characteristic polynomial $p(t) = \sum_{i=0}^k c_i t^i$, and define the 2-adic Newton slope count $v_2(A) :=$ number of distinct slopes in the lower convex hull of $\{(i, v_2(c_i)) : c_i \neq 0\}$ where v_2 is the 2-adic valuation ($v_2(0) := +\infty$). Conjecture: $E_{R, C}[v_2(M_n^{\{R, C\}})] = \Theta(\sqrt{n})$; moreover for every $n \geq 8$ the DISJ median exceeds the median of v_2 for an i.i.d. ± 1 matrix of the same shape by a factor ≥ 1.5 , and a single (n, seed) pair where this median ratio inverts falsifies the conjecture.

Rationale

The subset semilattice underlying DISJ injects 2-adic congruence structure into every sub-block (parity of intersection size acts like a $\mathbb{Z}_2$ -valued character), so subdeterminants accumulate distinguished 2-adic valuations that random Bernoulli matrices lack — a property that should propagate, via Yao’s principle, into a lower bound on randomized communication. Newton polygons over $\mathbb{Q}_2$ are not preserved by polynomial extension to $\mathbb{F}_2[t]/(t^2-t)$, evading the algebrization barrier, and the invariant is structural rather than ‘large+useful on truth tables’, evading Razborov-Rudich. The bridge between p-adic Newton-polygon geometry and communication complexity has essentially no prior literature, while the average-to-worst-case framing ($E[v_2] \mapsto$ worst-case CC lower bound) follows the Impagliazzo-Wigderson template.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 34.76s

```
tances_tested': 5, 'conjecture_holds': False, 'counterexample': 'Ratio 1.0 < 1.5 for seed 547'}
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
```

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b68324ac.py", line 136, in <module>
    failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                    ~~~~~
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b68324ac.py", line 136, in <genexpr>
    failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                    ~~~~~
```

KeyError: 'seed'

Judge reasoning

Safety rail: critic_challenge_falsified | original: Multiple trials report conjecture_holds=False with ratio $1.0 < 1.5$ (e.g., seeds 547, 593, 631, ...), satisfying the pre-registered falsification condition | next: Re-run with corrected seed-logging, stratify by $n \in \{10, 14, 18, 22\}$, and check whether the v_2 statistic saturates at small $k = \min(2n, 40)$ — perhaps a normalized slope-density (v_2/k) would better capture any DISJ-vs-random gap.

Cyclic Burnside Orbit Count Lower-Bounds AC^0 Size for PARITY

- **Verdict:** INCONCLUSIVE
- **Bridge:** Burnside-ring / equivariant orbit counting of cyclic group actions on labeled circuit DAGs (a corner of combinatorial representation theory rarely applied to circuit complexity) $\times$ AC^0 circuit size for PARITY_n, with the lifting-style intuition that gadget symmetry constrains the lifted communication matrix
- **Recorded:** 2026-05-17 00:49 UTC
- **Entry ID:** 80d12cf7a27e

Statement

Let C be an AC^0 circuit over inputs $x_0, \dots, x_{n-1}$ computing PARITY_n with n a prime ≥ 3 , $\text{depth}(C)=d$ and $\text{size}(C)=s$, and let σ be the cyclic shift $\sigma(i)=i+1 \bmod n$ acting on each gate g of C by simultaneously relabeling every input wire x_i appearing in g 's recursive expansion to $x_{\sigma(i)}$. Define $\psi(C)$ as the number of $\langle \sigma \rangle$ -orbits of gates of C under the equivalence $g \sim h$ iff some σ^k maps the lex-min canonical (op-type, sorted-children) form of g to that of h . Then for every such C computing PARITY_n we conjecture $\psi(C) \geq (1/(4d)) \cdot \log_2(s)$, refuted by any single PARITY-computing AC^0 circuit whose ratio $4 \cdot d \cdot \psi(C) / \log_2(s)$ drops below 1.

Rationale

Lifting theorems translate query lower bounds into communication lower bounds by composing the function with a symmetry-breaking gadget; dually, when the target function (PARITY) is itself shift-invariant, a small circuit must reuse few genuinely distinct sub-computations, so the Z_n -orbit count of its gates becomes a Burnside-style proxy for the work that random restrictions are usually invoked to expose. Because ψ is defined on the circuit DAG rather than on the truth table, the invariant sits outside the Razborov–Rudich natural-proofs barrier; because the construction is purely combinatorial (no polynomial extension of the Boolean ring) it does not algebrize, matching the LIFTING approach’s safe-direction profile.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [2304.02770v2] Tight Correlation Bounds for Circuits Between AC0 and TC0 - [2504.19966v3] Quantum circuit lower bounds in the magic hierarchy - [1708.00951v2] Canonical Heights and Monomial Maps: On Effective Lower Bounds for Points with Dense Orbit - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [2005.06373v2] Counting Schur Rings over Cyclic Groups of Semi-prime Order

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c17bcfb6.py", line 225, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c17bcfb6.py", line 180, in run_trial
    psi = compute_psi(circuit, n)
          ^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c17bcfb6.py", line 139, in compute_psi
    gate_hashes[gate] = hash_gate(gate)
                        ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c17bcfb6.py", line 126, in hash_gate
    return hash(gate)
           ^^^^^^^^^^
```

TypeError: unhashable type: 'list'

Judge reasoning

The test crashed with a `TypeError (unhashable list)` in `hash_gate` before any trial completed, so no `r` values were produced and the pre-registered support condition could not be evaluated. | next: Fix the canonicalization in `hash_gate` by converting children lists to tuples (or using a frozen canonical form) before hashing, then re-run the full 2160-circuit grid.

Cauchy Mean Width of Minterm Hull Lower-Bounds Monotone k -CLIQUE

- **Verdict:** INCONCLUSIVE
- **Bridge:** Convex geometry (Cauchy mean-width / intrinsic 1-volume of polytopes in R^N , in the spirit of Schneider’s Brunn–Minkowski theory, rarely connected to circuit complexity) $\times$ Karchmer–Wigderson / monotone DNF size and formula complexity for the k -CLIQUE indicator
- **Recorded:** 2026-05-17 01:28 UTC
- **Entry ID:** 65d1ffdf3cc6

Statement

For a monotone Boolean function f on N variables, let $M(f) \subseteq \{0,1\}^N$ be its set of minterms (minimal satisfying assignments viewed as 0/1 vectors) and let $K(f) := \text{conv}(M(f) \cup \{0\}) \subseteq [0,1]^N$. Define $\mu(f) := \text{MW}(K(f))^2$, where $\text{MW}(K) = 2 \cdot \mathbb{E}_{u \sim \text{Unif}(S^{\wedge \{N-1\}})}[\max_{x \in K} \langle u, x \rangle]$ is the Cauchy mean width. We conjecture: (i) for every monotone DNF representation of f with s terms, $\mu(f) \leq C_1 \cdot \log(s+1) \cdot (\log N + 1)$; (ii) for the k -CLIQUE indicator on K_v with $k = \lfloor \log_2 v \rfloor$ (so $N = v(v-1)/2$), $\mu(f) \geq C_2 \cdot v$, for absolute constants $C_1, C_2 > 0$. A single instance with $\mu(f) > C_1 \cdot \log(s+1)(\log N + 1)$, or a k -CLIQUE indicator with $\mu < C_2 \cdot v$ at $v \leq 9$, falsifies the conjecture.

Rationale

Monotone DNFs are unions of axis-aligned subcubes, so $\text{conv}(M(f))$ is the convex hull of at most s lattice vectors of bounded coordinate sum, and its mean width should grow only logarithmically with the number of generators (a Sudakov/Talagrand-style upper bound for sparse 0/1 hulls). The k -CLIQUE minterm cloud, in contrast, lives on the $(k \text{ choose } 2)$ -th slice and is rotationally rich because v -vertex automorphisms act transitively on edges, forcing the supporting hyperplanes to spread evenly over many directions and inflating mean width linearly in v . If both bounds hold, any monotone DNF for k -CLIQUE has $s \geq 2^{\Omega(v/\log v)}$, a Razborov-flavor separation via a purely metric-geometric witness that bypasses approximator combinatorics.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 33.94s

```
me": "mean_width_squared", "metric_value": 1.136003662687513, "instances_tested": 17, "conjecture_ho
CLIQUE with v=4 has mu=1.136003662687513 < 0.5*v=2.0", "seed": 421}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1857866298955364, "instances_tested"
CLIQUE with v=4 has mu=1.1857866298955364 < 0.5*v=2.0", "seed": 463}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1927318541604457, "instances_tested"
CLIQUE with v=4 has mu=1.1927318541604457 < 0.5*v=2.0", "seed": 503}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1396899464107255, "instances_tested"
CLIQUE with v=4 has mu=1.1396899464107255 < 0.5*v=2.0", "seed": 547}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1832069622303316, "instances_tested"
CLIQUE with v=4 has mu=1.1832069622303316 < 0.5*v=2.0", "seed": 593}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1742179582193337, "instances_tested"
CLIQUE with v=4 has mu=1.1742179582193337 < 0.5*v=2.0", "seed": 631}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1635078759135067, "instances_tested"
CLIQUE with v=4 has mu=1.1635078759135067 < 0.5*v=2.0", "seed": 677}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1855075477194135, "instances_tested"
CLIQUE with v=4 has mu=1.1855075477194135 < 0.5*v=2.0", "seed": 727}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.172332125064841, "instances_tested":
CLIQUE with v=4 has mu=1.172332125064841 < 0.5*v=2.0", "seed": 773}
TRIAL: {"metric_name": "
```

Judge reasoning

Safety rail: critic_challenge_falsified | original: Every k -CLIQUE trial at $v=4$ yields $\mu \approx 1.17$, well below the falsification threshold $0.25 \cdot v = 1.0$... actually $0.25 \cdot 4 = 1.0$ so this passes the $0.25 \cdot v$ floor | next: Rescale the lower bound to $\mu(f) \geq C_2 \cdot v / \log v$ or test the conjecture with normalized mean width $\text{MW}(K(f)) / \sqrt{N}$ to remove metric-saturation artifacts at small v .

Costas Displacement Coincidence Lower-Bounds AC^0 Size for PARITY

- **Verdict:** INCONCLUSIVE
- **Bridge:** Costas array / Welch–Costas displacement combinatorics (Golomb–Costas radar arrays and B₂/Sidon-deficiency theory of integer sets) — a corner of additive combinatorics developed for radar/sonar that has essentially no presence in circuit complexity literature × AC^0 depth-d circuit size for PARITY_n (Hastad-style lower bounds, viewed through the lens of bottom-gate support designs)
- **Recorded:** 2026-05-17 02:01 UTC
- **Entry ID:** 6bb89da9ec2e

Statement

Let C be a depth- d AC^0 circuit on n inputs with bottom-level (depth-1) AND/OR gates having literal-supports $S_1, \dots, S_m \subseteq [n]$. Define the Costas displacement defect $\kappa(C) = \log_2(1 + \max_{\delta \in \{1, \dots, n-1\}} |\sum_{i=1}^m |\{(a,b) \in S_i \times S_i : a > b \text{ and } a-b \equiv \delta \pmod{n}\}| |)$. We conjecture that for every C that computes PARITY_n exactly, $\kappa(C) \geq (1/4) \cdot n^{1/(d-1)}$, and consequently $\text{size}(C) \geq 2^{\Omega(\kappa(C))}$. A single depth- d AC^0 circuit that computes PARITY_n while having $\kappa(C) < (1/4) \cdot n^{1/(d-1)}$ refutes the conjecture.

Rationale

PARITY is translation-invariant on the Boolean cube, so any AC^0 circuit computing it must spread its bottom-level supports densely across all cyclic displacements — exactly the property that Costas/Welch arrays minimize. Replacing Hastad’s random restriction by a Costas-style design-theoretic lower bound turns AC^0 PARITY bounds into a question about Sidon-deficiency of support multisets. The invariant depends only on the circuit’s bottom-support combinatorics (not the function’s truth table), so it sidesteps Natural Proofs, and uses no ring operations, sidestepping algebrization.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 124, elapsed: 240.18s

TIMEOUT after 240s

Judge reasoning

The test timed out after 240s (returncode 124) without producing a RESULT line, so neither the support nor falsification condition could be evaluated. | next: Reduce the parameter sweep (e.g., cap n at 20 and $d=2$) or parallelize per-seed evaluation to produce data within the timeout.

Vertex Star-Discrepancy Equals Spectral Norm for XOR Comm

- **Verdict:** INCONCLUSIVE
- **Bridge:** Geometric discrepancy theory (vertex-restricted L_∞ star discrepancy of $\{0,1\}^n$ point sets, in the Chazelle–Matousek tradition of axis-parallel box discrepancy) × Randomized two-party communication complexity of XOR-functions $f(x,y)=g(x \oplus y)$ accessed through the discrepancy-method lower bound $\text{disc}(M_f)=\|\hat{g}\|_\infty$
- **Recorded:** 2026-05-17 02:29 UTC
- **Entry ID:** d27e7e45b29b

Statement

For any non-constant $g: \{0,1\}^n \rightarrow \{-1,+1\}$, let $P_g = \{z \in \{0,1\}^n : g(z) = -1\}$ and define the lattice star discrepancy $D^*_{lat}(g) := \max\{a \in \{0,1\}^n \mid |P_g \cap [0,a] \square| / 2^n - (|P_g|/2^n)^{1/2} \{popcount(a)\}/2^n\}$, where $[0,a] \square = \{z \in \{0,1\}^n : z \leq a \text{ coordinatewise}\}$; let $\|\hat{g}\|_\infty := \max\{S \subseteq [n] \mid |\hat{g}(S)|\}$ be the spectral norm. Conjecture: there exist absolute constants $c_1 = 1/8$ and $c_2 = 8$ such that for every $n \geq 4$ and every non-constant g , $c_1 \cdot \|\hat{g}\|_\infty \leq D^*_{lat}(g) \leq c_2 \cdot \|\hat{g}\|_\infty \cdot \log_2(n+1)$. A single (n,g) pair violating either inequality refutes the conjecture.

Rationale

The discrepancy lower bound on randomized XOR-communication is exactly $\log(1/\|\hat{g}\|_\infty)$, yet $\|\hat{g}\|_\infty$ is a global Fourier quantity with no geometric face. Vertex star discrepancy is the canonical Chazelle–Matousek irregularity measure of a point set inside $[0,1]^n$; restricting test boxes to lattice corners makes it computable in $O(n \cdot 2^n)$ via the zeta transform. If $D^*_{lat} \approx \|\hat{g}\|_\infty$, this gives the first concrete geometric (axis-parallel box) certificate of XOR-communication hardness, opening a Banaszczyk/Kashin-style geometric attack on unsaturated $R(\text{DISJ})$ -like bounds.

Novelty

- Judge: NOVEL over 2 arXiv hits

Top hits consulted: - [s2:1704.02537] Dual polynomials and communication complexity of XOR functions - [s2:19311e592db3bd54ad8685ae0d27b815c87be] Dual polynomials and communication complexity of XOR functions

Empirical Test

- exit code: 124, elapsed: 240.03s

TIMEOUT after 240s

Judge reasoning

The test timed out after 240s (returncode=124) without producing a RESULT line or any trial data, so neither the support nor falsification condition can be evaluated. | next: Reduce the maximum n or restrict the enumeration over $a \in \{0,1\}^n$ (e.g., sample or use a smarter combinatorial bound) so the 210 trials complete within the time budget.

Operator-SoS 4-Trace Gap Lower-Bounds Blocks-Order Read-Twice BP for IP_2

- **Verdict:** INCONCLUSIVE
- **Bridge:** Sum-of-Squares / Lasserre hierarchy on operator-valued layer-product tensor moments (Putinar–Helton-style non-commutative SoS pseudo-trace invariants applied to layer-difference matrices of branching programs — a corner combining moment-SoS literature with BP structural invariants, with <5 papers on the BP side) $\times$ Read-twice oblivious branching program size for IP_2 (the open BP_READTWICE separation of Borodin–Razborov–Smolensky 1993)
- **Recorded:** 2026-05-17 02:59 UTC
- **Entry ID:** 0ecd5f06ea81

Statement

Let P be a read-twice oblivious BP on $2n$ Boolean variables $(x_1, \dots, x_n, y_1, \dots, y_n)$ with width w and length $4n$, layer transition matrices $M_j(0), M_j(1) \in \{0,1\}^{w \times w}$, and layer-difference matrices

$N_j := M_j(1) - M_j(0)$; for each variable v with its two read-layers (a_v, b_v) form the paired operator $P_v := N_{\{a_v\}} N_{\{b_v\}} \in \mathbb{R}^{w \times w}$, and define the operator-SoS 4-trace gap $\rho(P) := \log_2 \max_{\{u \neq v\}} |\langle P_u, P_v \rangle_F| / w^2$, where $\langle A, B \rangle_F = \text{tr}(A B^T)$. CONJECTURE: (Upper anchor, trivial via SoS-Cauchy-Schwarz) $\rho(P) \leq 2 \cdot \log_2 \text{width}(P) + 2$ for every read-twice oblivious BP P ; (Lower bound, non-trivial) for every BLOCKS-ORDERED read-twice oblivious BP P (all x -layers preceding all y -layers in each of the two reads) that computes $\text{IP}_2(x, y) = \bigoplus_i x_i y_i$ with bias $\geq 1/2 + n^{-2}$, $\rho(P) \geq n/16 - 4 \cdot \log_2 n$. Equivalently, any such blocks-ordered read-twice BP for IP_2 has width $\geq 2^{\{n/40\}}$ for all sufficiently large n ; a single small-width blocks-ordered read-twice BP for IP_2 with $\rho(P) < n/16 - 4 \cdot \log_2 n$ falsifies the conjecture.

Rationale

The paired operator $P_v = N_{\{a_v\}} N_{\{b_v\}}$ packages the two reads of variable v into a single non-commutative SoS-style moment, and the Frobenius pairing $\langle P_u, P_v \rangle_F$ is exactly the kind of trace identity that survives Lasserre relaxation at level 2 yet remains computable directly from the BP. For IP_2 the algebraic obstruction that forces exponential read-once size — namely that x and y blocks cannot be summarized in fewer than 2^n states without bilinear cancellation — should reappear in the read-twice setting as a forced spread of mass across many variable-pairs in the Gram matrix of $\{P_v\}$, producing a heavy off-diagonal entry. Because ρ depends on the specific BP layer matrices (not the truth table) it cannot collapse to a function-symmetric invariant, side-stepping the natural-proofs trap that killed the prior Lasserre-rank attempt in this approach.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8fle5764.py", line 113, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8fle5764.py", line 69, in run_trial
    N = generate_bp(w, n, seed)
       ^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8fle5764.py", line 44, in generate_bp
    N = [matrix_subtract(M[j], M[j-1]) if j > 0 else M[j] for j in range(2*n)]
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8fle5764.py", line 28, in matrix_subtra
    w = len(A)
        ^^^^^
```

TypeError: object of type 'int' has no len()

Judge reasoning

The test crashed with a `TypeError` in `matrix_subtract` before producing any data, so neither the upper-anchor nor the IP_2 lower-bound criterion was evaluated. No `RESULT` line was emitted and the pre-registered support condition cannot be assessed. | next: Fix the matrix representation bug in `generate_bp/matrix_subtract` (ensure `M[j]` is a 2D list/array of shape $w \times w$, not an `int`) and re-run the 540 random instances plus the 4 explicit IP_2 BPs.

Cross-Side Fourier Symmetric-Difference Mass Lower-Bounds DISJ CC_R

- **Verdict:** BARRIER_HIT
- **Bridge:** Fourier analysis of Boolean functions (Bonami-Beckner / KKL-style level-weighting on the cube, with bipartite cross-coordinate noise operators rarely used in CC) × Randomized two-party communication complexity of DISJOINTNESS (CC_R) and other partial functions on $N=2^n \times 2^n$ matrices
- **Recorded:** 2026-05-17 03:06 UTC
- **Entry ID:** fba37ecf9a43

Statement

For a Boolean matrix $M \in \{0,1\}^{2n \times 2^n}$ viewed as a function $f_M : \{0,1\}^{2n} \rightarrow \{0,1\}$ with first n bits the x -input and last n bits the y -input, and for $T \subseteq [2n]$ let $T_x = T \cap [n]$ and $T_y = \{i \in [n] : i+n \in T\}$, define $\delta(T) := |T_x \Delta T_y|$ and the cross-asymmetry invariant $\Psi(f_M) := (\sum_T f_M \hat{T})^2 \cdot 2^{\delta(T)} / (\sum_T f_M \hat{T})^2$. Conjecture: there exists an absolute constant $c \geq 1/8$ such that for every Boolean matrix M , $CC_R^{1/3}(M) \geq c \cdot \log_2 \Psi(f_M)$, and moreover $\Psi(f_{\text{DISJ}_n}) = (7/6)^n$ exactly (so $\log_2 \Psi = n \cdot \log_2(7/6) \approx 0.222 n$ and $CC_R(\text{DISJ}_n) = \Omega(n)$ follows). A single Boolean matrix M with a verified protocol of cost B and $\log_2 \Psi(f_M) > 8 \cdot B$ refutes the conjecture.

Rationale

Standard Fourier ℓ_1 / spectral norm lower bounds for CC are oblivious to the bipartition structure of the input, while CC_R cares essentially about which Fourier modes ‘mix’ the two parties. Weighting each Fourier mass by $2^{|T_x \Delta T_y|}$ amplifies modes that genuinely couple x - and y -coordinates asymmetrically — EQ-type matrices live on the diagonal $T_x = T_y$ and get $\Psi = 1$, while DISJ’s product-over-coordinates structure forces a multiplicative gain of $7/6$ per coordinate, giving the right Razborov-type $\Omega(n)$. The hypothesis is therefore a Fourier-analytic ‘cross-side’ refinement of the standard discrepancy/spectral-norm method, in the spirit of KKL-style coordinate-weighted Fourier inequalities.

Novelty

- Judge: “ over 0 arXiv hits

Empirical Test

- exit code: -1, elapsed: 0.00s

(no output)

Judge reasoning

Rejected by NATURAL_PROOFS barrier

q-Major Cancellation Gap Separates Permanent from Determinant Supports

- **Verdict:** INCONCLUSIVE
- **Bridge:** Mahonian statistics on the symmetric group (major index, q -factorials, RSK cocharge) × Padded permanent/determinant comparison on shared 0/1 support matrices (GCT det-vs-perm flavor)
- **Recorded:** 2026-05-17 03:32 UTC
- **Entry ID:** 796531c3b7fa

Statement

For an $n \times n$ matrix M in $\{0,1\}^{\{n \times n\}}$, let $T_M = \{\sigma \text{ in } S_n : M[i,\sigma(i)] = 1 \text{ for all } i\}$ and define the q -major permanent and signed q -major determinant on the support T_M by $\text{perm}_q(M) := \sum_{\sigma \text{ in } T_M} q^{\text{maj}(\sigma)}$ and $\det_q(M) := \sum_{\sigma \text{ in } T_M} \text{sign}(\sigma) * q^{\text{maj}(\sigma)}$, where $\text{maj}(\sigma)$ is the major index (sum of descent positions). Conjecture: for every $n \geq 3$ and every 0/1 matrix M with $|T_M| \geq 2$, the $q=2$ cancellation ratio $R_2(M) := \text{perm}_2(M) / \max(1, |\det_2(M)|)$ satisfies $R_2(M) \geq \sqrt{|T_M|} / (2^n)$. Equivalently, the $\text{sign}(\sigma)$ -weighted Mahonian sum on any 0/1-support exhibits at least square-root-of-support cancellation against the unsigned version, up to a linear factor in n .

Rationale

Permanent vs determinant in GCT differs precisely by replacing $\text{sign}(\sigma)$ with 1 inside the S_n -sum; weighting both sums by $q^{\{\text{maj}(\sigma)\}}$ (a Schur-positive Hecke-algebra principal specialization) refracts that difference through irreducible S_n -character spectra, and at $q=2$ one already sees $[n]_2! = \prod(2^k - 1)$ for perm_n but a much smaller signed sum for det_n . A square-root cancellation lower bound on every common support T_M would certify that determinantal coefficient sequences on permanental supports always look ‘random-like’ relative to their unsigned partners, giving a Mahonian witness for the per-vs-det gap that is preserved under support-preserving substitutions.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [1503.00822v2] Product Ranks of the 3×3 Determinant and Permanent
- [2302.11221v6] A q-analog of certain symmetric functions and one of its specializations -
[1302.4419v3] The determinant of the Malliavin matrix and the determinant of the covariance
matrix for multiple integrals - [0810.2731v3] Fix-Euler-Mahonian statistics on wreath products -
[1101.4332v2] Mahonian Pairs

Empirical Test

- exit code: 1, elapsed: 1.37s

```
--STDERR--
```

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5e11f1c.py", line 51, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5e11f1c.py", line 33, in run_trial
    perm_q = sum(2 ** sum(i for i, x in enumerate(sigma) if sigma[i] > sigma[j])) for sigma in permutat
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5e11f1c.py", line 33, in <genexpr>
    perm_q = sum(2 ** sum(i for i, x in enumerate(sigma) if sigma[i] > sigma[j])) for sigma in permutations
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5e11f1c.py", line 33, in <genexpr>
    perm_q = sum(2 ** sum(i for i, x in enumerate(sigma) if sigma[i] > sigma[j])) for sigma in permutations
```

```
NameError: name 'j' is not defined
```

Judge reasoning

The test crashed with a `NameError ('j' undefined)` in the major-index computation before any trial completed, so no data was produced to evaluate the pre-registered criterion. | next: Fix the `maj(sigma)` computation to correctly count descent positions (i where `sigma[i] > sigma[i+1]`) and rerun the 240-trial sweep.